

```
In [1]: import pandas as pd
import numpy as np
from collections import Counter
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df=pd.read_csv('spotify_long_tracks_2014_2024.csv')
print(f"columns\t:{df.shape[1]}\nrows\t:{df.shape[0]}")
```

```
columns :4
rows    :816
```

```
In [3]: df.head()
```

```
Out[3]:
```

	ID	Name	Duration (Minutes)	Artists
0	026O2YP2wRI9h3Ht9ClZST	Steady Rain in a Forest with Light Background ...	100	Nature Sounds
1	0DXnjJxJBvgMvBjG6Jug7I	Soundarya Lahari	66	Mambalam Sisters
2	6BFn78qFy9bRqKHTMfhNv	Waves of Abundance & Fullfillment	60	Zen Life Relax
3	1dNwMOMtq3yflXibbqgLku	Mosquito Solution Anti Teen Street Party or Te...	45	Device Sound Effect
4	290gfei8moWWSh8qTVkTT3	Ibiza Sunset Tibetan Singing Bowl Sessions 7 (...	44	Tibetan Singing Bowls

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 816 entries, 0 to 815
Data columns (total 4 columns):
 #   Column                Non-Null Count  Dtype
---  -
 0   ID                    816 non-null   object
 1   Name                  816 non-null   object
 2   Duration (Minutes)    816 non-null   int64
 3   Artists               816 non-null   object
dtypes: int64(1), object(3)
memory usage: 25.6+ KB
```

```
In [5]: # cleaning data
df.columns=df.columns.str.lower().str.replace(' ','_')
```

```
In [6]: # cek data unique
for column in df.columns.tolist():
    print(column)
    print (len(df[column].unique ()))
```

```
id
816
name
816
duration_(minutes)
29
artists
812
```

```
In [7]: count=Counter(df['artists'])
        print (count)
```

Counter({'Nusrat Fateh Ali Khan': 2, 'Tangerine Dream': 2, 'Joachim Pastor': 2, 'Suara': 2, 'Nature Sounds': 1, 'Mambalam Sisters': 1, 'Zen Life Relax': 1, 'Device Sound Effect': 1, 'Tibetan Singing Bowls': 1, 'Gontaro Yanagiya': 1, 'Pēteris Vasks, O/Modernt Chamber Orchestra, Hugo Ticciati': 1, 'Jessica Cross': 1, 'Relaxmydog': 1, "Nicole Lizée, Canada's National Arts Centre Orchestra, Alexander Shelley": 1, 'LOKOS TV Now Meditation Music': 1, 'Len Leise': 1, 'Sapien Medicine': 1, 'Al Sheikh Ahmad Alnufais': 1, 'Utamaru Katsura': 1, 'Sun Ra': 1, 'Felipe Pigna, Roberto Martínez': 1, 'Edres Abkar': 1, 'Satorio': 1, 'Emmett Miller, Steven Halpern': 1, 'Blue Claw Fitness': 1, 'Tha Lero': 1, 'Anton Bruckner, Bavarian Radio Symphony Orchestra, Otto Klemperer': 1, 'Krishna Das, Sharon Salzberg': 1, 'Katie Weatherup': 1, 'Franz Schmidt, Wiener Philharmoniker, Zubin Mehta': 1, 'S.E.T.I.': 1, 'Luciano Berio, Institute for Computer Music and Sound Technology, Georg Kohler': 1, 'Lucky Star no Minna': 1, 'Ionela Pascu': 1, 'Küçük Murat': 1, 'Johann Nepomuk Hummel, Dieter Klöcker, Florian Uhlig, Consortium Classicum': 1, 'Ludwig van Beethoven, Berliner Philharmoniker, Herbert von Karajan': 1, 'Hatem Al Iraqi': 1, 'Lucas King': 1, 'Richard Wagner, Stuttgart Radio Symphony Orchestra, Hans Müller-Kray': 1, 'Herman Basudde': 1, 'Anton Bruckner, Wiener Philharmoniker, Herbert von Karajan': 1, 'Hans Pfitzner, Rundfunk-Sinfonieorchester Berlin, Alban Gerhardt, Sebastian Weigle': 1, 'Allegaeon': 1, 'Antonín Dvořák, Alisa Weilerstein, Czech Philharmonic, Jiří Bělohlávek': 1, 'Dimitri Murrath, Vincent Planès': 1, 'Win Oo': 1, 'Dmitry Kabalevsky, Torleif Thedéen, Hannover Radio Philharmonic Orchestra, Eiji Oue': 1, 'Ludwig van Beethoven, The Beethoven Quartet': 1, 'مكرم المندوب': 1, 'Bvdub': 1, '妙蓮華': 1, 'Kaipa': 1, 'Skinnerbox': 1, 'Sweet Carol': 1, 'Zaïko Langa Langa, Nkolo Mboka': 1, 'King of the World': 1, 'Frédéric Chopin, Folke Nauta': 1, 'Cheb Hichem': 1, 'Macroform': 1, 'Howard Givens, Craig Padilla': 1, 'Dayspring Worship, Joel Wiseman': 1, 'Wolfgang Amadeus Mozart, Les Vents Français': 1, 'Musarra': 1, 'Nora del Mar': 1, 'Relaxing888': 1, 'Arabic Djs, Arabic DJ': 1, 'Ludwig van Beethoven, Ji Hae Park, Simon Lepper': 1, 'Barry Jamieson': 1, 'Kelly Andrew': 1, 'Wolfgang Amadeus Mozart, Eduardo Marturet, Emmy Verhey, Royal Concertgebouw Orchestra': 1, 'Oscar Navarro Gonzalez, Oscar Navarro, Oscar Navarro Symphony Orchestra': 1, 'Endless Ocean': 1, 'New Members': 1, 'Lee Weisert, Clara Yang, Juan Alamo': 1, 'Alberto Ginastera, Donald Sulzen, Henry Raudales': 1, 'Jorgio Kioris, Simos Tagias, Hot Tuneik': 1, '1 Mile North': 1, 'Black Foxxes': 1, 'Z-Arc': 1, 'Carl Michael Ziehrer, Original C.M.Ziehrer Orchester, Hans Schadenbauer': 1, 'George Banton': 1, 'Mourners Lament': 1, 'Loom, Dopadocks': 1, 'Invisible Reality, Klopffeister': 1, 'Derya Türkan, Sokratis Sinopoulos': 1, 'Bongo & Pusk': 1, 'Dren McDonald': 1, 'Cornershop': 1, 'Glaubenszentrum Live, Markus Muntean': 1, "Claude Debussy, Orchestra dell'Accademia Nazionale di Santa Cecilia, Leonard Bernstein": 1, 'Dziviva': 1, 'Vous Autres': 1, 'Alienn': 1, 'Trey Anastasio': 1, 'Alexa Sunshine Rose, Shmishai': 1, 'Cyril': 1, 'Andawan': 1, 'Sanam Marvi': 1, 'Drive-By Truckers': 1, 'O.C., Fat Joe, Lord Finesse, Big L, A.G., Diamond D': 1, 'Mindwave': 1, 'Marimba Orquesta La Gran Manzana': 1, 'yeahyoutoo': 1, 'Dyce': 1, 'Kyle Bobby Dunn': 1, 'Causa Sui': 1, 'Ali Krasniqi, Halil Bytyqi': 1, 'Jean Sibelius, Turku Philharmonic Orchestra, Leif Segerstam': 1, 'Alpha Gravity': 1, 'Osvit': 1, 'Matthew Fries, Phil Palombi': 1, 'Orquesta Amor Latino': 1, 'KHAIKHAN, Dest': 1, 'Simon Le Grec': 1, 'Mate Bulic': 1, 'Andres Sanchez, Katherine Amy': 1, 'Ivan Graziani': 1, 'Antlers': 1, 'Pspiralife': 1, 'Metallica': 1, 'Hossam Ramzy': 1, 'Pascal Voggenhuber': 1, 'Jacky Mingo, Kapote': 1, 'Antonín Dvořák, Northern String Quartet': 1, 'Crazy P': 1, 'Harinama Ruci': 1, 'Susheela Raman': 1, 'Gazal Alabedallah': 1, 'The Mambalams Sisters': 1, 'Federico Scavo': 1, 'Ricardo Piedra, Kay-D': 1, 'Neptune Project, Polly Strange, The Noble Six': 1, 'Antonín Dvořák, Budapest Philharmonic Orchestra, Stephen Pomerantz': 1, 'Pyotr Ilyich Tchaikovsky, Philharmonia Orchestra, Sir Charles Mackerras': 1, 'John Tavener, Andrew Lumsden, Winchester Cathedral Choir, George Castle': 1, 'Miguel Zenón': 1, 'Jimmy Dawkins': 1, 'Tommy Castro': 1, 'Mike Casey': 1, 'Ralph Myerz, R.E.C., Ryan

James, SUPER SPACE PROJECT': 1, 'Róisín Murphy, Crooked Man': 1, 'CHANDAN SHARMA HATHRASI': 1, 'Depeche Mode': 1, 'Héritier Wata': 1, 'Spencer Parke r': 1, 'Andruss': 1, 'FELIZ': 1, 'Solarc, Criss Source': 1, 'Funk You': 1, 'Kamilo Sanclemente': 1, 'Love From San Francisco': 1, 'Traumton': 1, 'Cas imir Von Oettingen': 1, 'ヒーリング・ライフ': 1, 'Shai T': 1, 'Rhyton': 1, 'U ttam Kumar Mondal': 1, 'Perception': 1, 'Chromeo, Toro y Moi': 1, 'Boy Nex t Door': 1, 'Connie Harvey': 1, 'Andres Blows': 1, 'Sonu Sargam Yadav': 1, 'Seol': 1, 'Dominika Gurbaľová': 1, 'North Coast Vibes, Rob G': 1, 'Cristo ph': 1, 'Wolfgang Amadeus Mozart, Anna Maunther, Tatrai Quartet': 1, 'Pabl o Held Trio, Pablo Held, Robert Landfermann, Jonas Burgwinkel': 1, 'Alessa ndro Penna': 1, 'Golden Era Records, Hilltop Hoods, Funkoars, A.B. Origina l, Briggs, Vents, K21, Purpose': 1, 'Edvard Grieg, Mikhail Pletnev': 1, 'S hahrokh': 1, 'Pragmatix': 1, 'Antonio Venuti & Mr Goaty': 1, 'Kayros, João Oliveira': 1, 'Phase 2': 1, 'Grateful Dead': 1, 'DJ Tarkan': 1, 'Alfoa': 1 , 'Stefano Noferini, Looad': 1, 'Jason Bye': 1, 'Ben DJ': 1, 'Tony Anderso n': 1, 'Max Freegrant, Max Meyer, DJ San': 1, 'Frank Donner, George Morel, Stefan Mint': 1, 'Birdcage, Avus': 1, 'Lag': 1, 'Simina Grigoriu': 1, 'Tra ditional, Gulshan Tandon': 1, 'Big Mojo': 1, 'Covenant Worship': 1, 'Dirty Culture, Nusha': 1, 'Dj Marta, Tim Wokan': 1, 'Jose Solano, Cafe De Anatol ia': 1, 'Bernd Kistenmacher': 1, "Gov't Mule": 1, 'SKYGO': 1, 'Limpodisco, James Rod': 1, 'Bender, Nina': 1, 'Valeron, 7even (GR)': 1, 'Elek3ctii': 1 , 'Wuta Mayi': 1, 'Cosmic Replicant': 1, 'Anna Lee': 1, 'Mijo': 1, 'Boris S.': 1, 'Massa (7)': 1, 'Noneoftheabove': 1, 'Cheb Housseem': 1, 'Butcher B rown': 1, 'Prince Rama': 1, 'Dj Fudalwokit': 1, 'Green Velvet, Jay Lumen': 1, 'Roger Spees': 1, 'Óperentzia': 1, 'Stereo Express': 1, 'bRUNA, Henry S aiz': 1, 'Simon Rackham': 1, 'Sincroniza2': 1, 'Kim Jens Witztenleiter, Wol fy-Office': 1, 'The Teknoist': 1, 'Matt Sassari': 1, 'Diana Krall': 1, 'Ma rk Michael': 1, 'Nick Magnus': 1, 'Mike Schommer': 1, 'Beatsole': 1, 'Mich ael Sweeney, Midwest Wind Ensemble': 1, 'Eddie Palmieri': 1, 'Matador': 1, 'Morgan Heritage': 1, 'Maiga': 1, 'Giovanni Battista Pergolesi, Blandine S taskiewicz, Les Ambassadeurs, Alexis Kossenko': 1, 'Matt Rowan': 1, 'T-SQU ARE': 1, 'The Cinematic Orchestra, Roots Manuva': 1, 'Pyotr Ilyich Tchaiko vsky, Eugene Ormandy, Philadelphia Orchestra': 1, 'Shafa Ullah Khan Rokhri ': 1, 'Nga Tae, Paddy Free': 1, 'Exz, Pigmalhão': 1, 'Nils Ohrmann': 1, 'G allery Six': 1, 'Danny Cruz, HP Vince': 1, 'Michal Prokop, Framus Five': 1 , 'Ava Irandoost': 1, 'LMI': 1, 'Laika & Strelka': 1, 'Sweet Little Band': 1, 'Lunatica': 1, 'Pacific State, Neil Curtis': 1, 'Ernesto Köhler, Denis Lupachev, Natalia Frolova': 1, 'Ceega Wa Meropa, Rony M, Petmuso': 1, 'Fil terheadz': 1, 'Alma Negra': 1, 'Agreen Dilshad': 1, 'Neurotribe': 1, 'Audi ophonic': 1, 'Maik Thomas': 1, 'Andres Blows, Pablo Say': 1, 'RezQ Sound': 1, 'Takuya Matsumoto': 1, 'Lightform, Zaa': 1, 'Pelle Gudmundsen-Holmgree n, Theatre Of Voices, Paul Hillier': 1, 'Devi Chitrlekha': 1, 'Maximiljan , Black Loops': 1, 'Ludwig van Beethoven, Martha Argerich, Julian Rachlin, Mischa Maisky': 1, 'Daniel Bortz': 1, 'Rene Amesz': 1, 'Matthew Lien': 1, 'Kooky, Damoon, Niki Kand': 1, 'Rothko': 1, 'Traditional, Virginia Band an d Orchestra Directors Association 2019 District XI All-District Concert Ba nd, Dr. Brendan Caldwell': 1, 'Karanyi': 1, 'Woo2Tech, Sensekraft': 1, 'Fe rnando Velázquez': 1, 'NMR Digital': 1, 'Neil Diamond': 1, 'European Jazz Trio': 1, 'Jamie Trench': 1, 'Libertad Maestro': 1, 'Tasya Rosmala': 1, 'S .P.Y': 1, 'SeamlessR': 1, 'Lewis Boardman': 1, 'Dj Darroo': 1, 'Olivier C ': 1, 'Healing Peace': 1, 'Art Department': 1, 'Ralph Kaminski': 1, 'Rubix Qube': 1, 'Messenger': 1, 'Tamas Skafar': 1, 'ພາບຮູບໂຄງ': 1, 'Borrowed Ide ntity': 1, 'Juan Pacifico, Rik-Art, Josephine Sweett, Bruno Kauffmann': 1, 'Los Kikers de Carora': 1, 'Furney': 1, 'Belako': 1, 'Omar Jr': 1, 'Abdull a Rashim': 1, 'Nexus': 1, 'The Cyclist Conspiracy': 1, 'Martin Merino, SOL DIERBEAT, Billy G': 1, '生祥樂隊': 1, 'Lea Cristina': 1, 'Answer Code Reque st': 1, 'Simple Symmetry': 1, 'Visage': 1, 'Trooper': 1, 'Michael Wong': 1 , 'Stradivarius': 1, 'Meditationsmusik Doktor': 1, 'Phonique, Lazarusman, Douglas Greed': 1, 'Lastraw, JazzyFunk': 1, 'Supacooks': 1, 'Ludwig van Be ethoven, Marek Janowski, WDR Sinfonieorchester Köln': 1, 'Mason Maynard':

1, 'Róisín Murphy': 1, 'Novalima': 1, 'Felo': 1, 'Sidekick, Daniele Petronelli, Worp': 1, 'Nora En Pure': 1, 'Mr. Pepper, So.young, Bu.Di': 1, 'Training Motivation Music': 1, 'PornoGraffiti': 1, 'Hijjaz': 1, 'natural/electronic.system.': 1, 'Paul Hindemith, Xi Zhai, Kronberg Academy Soloists, Schleswig-Holstein Festival Orchestra, Christoph Eschenbach': 1, 'Crystal Geometry': 1, 'DJ Nasty 2': 1, 'The Devil & the Universe': 1, 'Tamara Weber': 1, 'Ital, Aurora Halal': 1, 'Andy Parsons': 1, 'Ripe, Audiotree': 1, 'East End Dubs, Enzo Siragusa': 1, 'Pr Chandra Sekaran': 1, 'Güç Başar Güllü': 1, 'Baal Gopal, Rudhav Sharma': 1, 'Libny Kattapuram, Celine Shoji': 1, 'Humanature, j:logic, MC ASTRO': 1, 'Nigel Good': 1, 'Dakar, Tim Baresko': 1, 'Zikri': 1, 'Bela Shende': 1, 'amazarashi': 1, 'Midhu Vincent': 1, 'Uncovering Within': 1, 'Terraròss': 1, 'Manilla Road': 1, 'Masajes Spas, Mantra para Dormir': 1, 'Bomb Squad Unit': 1, 'Papi Sousa': 1, 'Dave Anthony, Beverlei Brown': 1, 'Madonna, Swae Lee, DJLW': 1, 'Varhat': 1, 'Karim Kamar': 1, 'Danniel selfmade': 1, 'Sweet Dreams': 1, 'Harris Jayaraj, Karthik': 1, 'Hirntot Posse, Dimi de la Ghetto': 1, 'Christian Ciotola': 1, 'Fine乐团': 1, 'GranDiosas, La Sonora Dinamita, Rocio Banquells, Dulce, Manolita Torres': 1, 'Popolo Vascio': 1, 'Camouflage': 1, 'Poschek': 1, 'Forest On Stasys': 1, 'Yash Chopra, Lata Mangeshkar, Sonu Nigam': 1, 'Sadistics': 1, 'Umwelt, The Exaltics': 1, 'Phonotronic': 1, 'Atjazz, Jullian Gomes': 1, 'Dennis Cruz, Iuliano Mambo': 1, 'Da Fresh, Max Freegrant, Slow Fish': 1, 'Antonello Venditti, Carlo Verdone': 1, 'Kei': 1, 'Luca Debonaire': 1, 'La Gossa Sorda': 1, 'The Protomen': 1, 'Tam Ca Áo Trắng': 1, 'FM de Zacapa': 1, 'Hans Zimmer, Rupert Gregson-Williams': 1, 'Heavenchord': 1, 'Om Meditation Bell': 1, 'Glitter, R3ckzet': 1, 'Yvonne Curtis': 1, 'The Unplugged Acoustics': 1, 'Nolek': 1, 'Nasta Super, Chris Cain': 1, 'Yoga World': 1, 'Annorkoth': 1, 'Bob Reynolds': 1, 'David Forbes, Richard Durand': 1, 'Bohemia Suburbana': 1, 'Anoraak': 1, 'Ofdrykkja': 1, 'Herr Lounge Corps, Cadaverous Condition': 1, 'Robert Broadway': 1, 'Don J Stephen': 1, 'Yaw Sarpong, Asomafo': 1, 'Traditional, Tamara Lukasheva, East Drive': 1, 'City': 1, 'Sub Love': 1, 'Eli Escobar, Nomi Ruiz': 1, 'Aaron Ahrends': 1, 'The Breathing Effect': 1, 'Jennifer Munday': 1, 'Author & Punisher': 1, 'Forever Philly Records, Quise Limye': 1, 'Kwartz': 1, 'Paul Jackson, Xantone Blacq, Tony Match': 1, 'Black Loops, Maik Yells': 1, 'Smooth Jazz Saxophone Band': 1, 'Polymod': 1, 'Esmerine': 1, 'Geila Zilkha': 1, "Parallax '48": 1, 'Eric Alexander Quartet': 1, 'Johnny Bass': 1, 'Paul De Silva': 1, 'w.y.k': 1, 'DJ Memo': 1, 'Dendera': 1, 'Davide Mazzilli': 1, 'Manu Kenton': 1, 'Kingdom Culture Worship': 1, 'Elkhorn': 1, 'Kizzmo': 1, 'Stiver': 1, 'Afrobeat, Tenisara, Nikitin': 1, 'Grand River': 1, 'Andy Hunter': 1, 'Streetslights': 1, 'Paloma San Basilio': 1, 'Embat Prod, Miki Leris, Alberto Casallo': 1, "DJ 'S'": 1, 'Ara Malikian, La Banda': 1, 'Ely Guerra': 1, '烏野薰爵士樂團': 1, 'Yury Ivánov': 1, 'Joseph Tivafire, The House of Blessings': 1, 'Chiomá Jesus': 1, 'Takuya Nakamura': 1, 'Anthony Mea': 1, 'Lady Zamar': 1, 'ASP': 1, 'Terrence Parker': 1, 'Gypsy and the Acid Queen': 1, 'Sorinel de la Plopeni': 1, 'Engin Titiz': 1, 'Lady Lizard': 1, 'Carter Vail': 1, 'Mama Schultz': 1, 'Krippsoulisc': 1, 'Stolas': 1, 'Franco Albertini': 1, 'Moonlight Jazz Blue': 1, 'Alex Haas': 1, 'Daze': 1, 'Tomohiko Sagae': 1, 'The Brecker Brothers': 1, 'Modd': 1, 'Tm Zaiko, Nuco, Zaiko y Nuco, Flaco, J Alexis, Griser Nsr, Hopper': 1, 'Auktyon': 1, 'Huminal': 1, 'Ross Couch': 1, 'Consecration': 1, 'Warren Haynes, Derek Trucks': 1, "Peru Inka's": 1, 'Maxim Vilarri': 1, 'Future Usses': 1, 'Iñaki Arakistain': 1, 'Rolling Contact': 1, 'Blair French': 1, 'Adryiano': 1, 'Dave Z': 1, 'Passion, Kristian Stanfill': 1, 'ALBATROS': 1, 'DJ CON-T': 1, 'Laura Taylor': 1, 'The Irish Rovers': 1, "Gabriel D'Or, Bordoy": 1, 'The New Division, Luke C habile': 1, 'Zach Winters': 1, 'LOUDNESS': 1, 'Al Sheikh Maher Al Muaiqly': 1, 'Lisa Lynne, Aryeh Frankfurter': 1, 'Remi Collins': 1, 'Joey Negro, Dave Lee, The Sunburst Band, Diane Charlemagne, Donna Gardier': 1, 'Sebastian Krieg': 1, 'Sarah McSweeney': 1, 'Andrea': 1, 'Wo Fat': 1, 'Cab Drivers': 1, 'Kery James, Anissa Stili': 1, 'Saphileaum': 1, 'Billy Esteban, Cafe De Anatolia': 1, 'Haydar Öztürk': 1, 'Yougaslava': 1, 'Mage': 1, 'Tino': 1, '

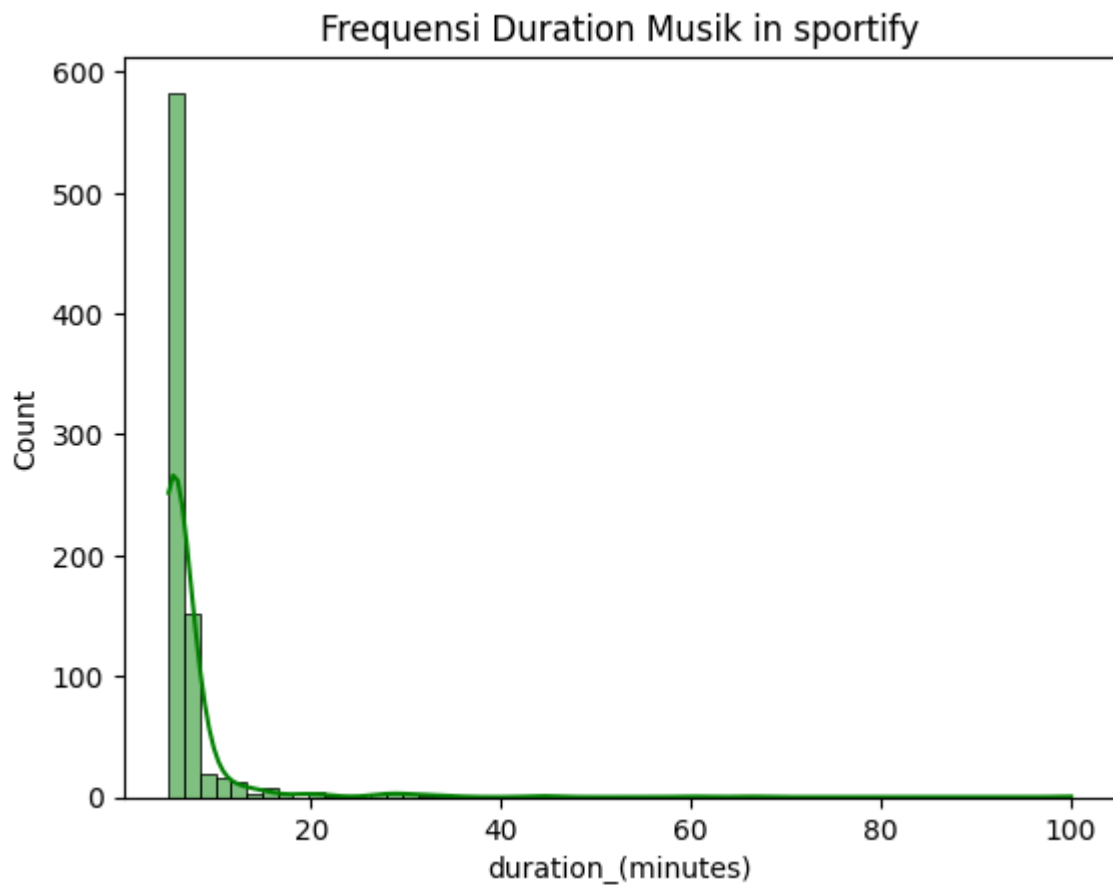
Harris Jayaraj, Dhanush, Marana Gana Viji, Naveen Madhav': 1, 'Sax Gordon': 1, 'Chicane': 1, 'The Transhumans': 1, 'Stamina': 1, 'Cindy Blackman Santana, John McLaughlin': 1, 'Race to Space': 1, 'Accept': 1, 'fhána': 1, 'S lam': 1, 'Nasteexo Indho': 1, 'Swami Tejomayananda': 1, 'Decadawn': 1, 'Keith Fortune, James Mac': 1, 'Young, Gladys': 1, 'The Silver Dimes': 1, '2814': 1, 'Joyita': 1, 'Timmy Regisford, Felicia Graham, Tiger Wilson': 1, 'Quentin Dujardin, Didier Laloy': 1, 'Hartmann': 1, 'Tài Nguyễn, Hoàng Anh': 1, 'Loud Harp': 1, 'Deadbeat UK': 1, 'Garrett David': 1, 'Ex Deo': 1, 'Edmund Hartsch, Ben Becker, Böhse Onkelz': 1, 'Chakra Dream': 1, 'Killbox': 1, 'The Juggernauts': 1, 'The Life Journey': 1, 'Lazarito Valdés y Bamboleo': 1, 'El Búho': 1, 'Planetself': 1, 'BLUEBETTO': 1, 'Paul Wilbur': 1, 'La Kritika': 1, 'Pet Music Academy': 1, 'Brandhård': 1, 'David Gravell, Ruby Prophet': 1, 'Replicant': 1, 'Skatebård, Lauer, Crabsticks, Igor Tipura': 1, 'Elfendy Tedja': 1, 'Pamungkas, The PeoplePeople': 1, 'SOUND HOLIC, Nana Takahashi': 1, 'Rosetta Stone': 1, 'Papa Grows Funk': 1, 'UMEMIRU': 1, 'globe': 1, 'Lika Morgan, C-Ro': 1, 'Void Vision': 1, 'FT-17': 1, 'Bungla': 1, 'Hamradun': 1, 'Lukas De Vrie': 1, 'Howling, Frank Wiedemann, RY X': 1, 'Andy Mientus': 1, 'Shakuntala Jadhav': 1, 'Tru Comers, Residentes, La Base': 1, 'Campursari Sangga Buana, Wulandari, Putri, Suji': 1, 'Roberto Choque, Romanza "La Magia del Sentimiento": 1, 'Sandi Thom': 1, 'LAMP IN TERREN': 1, 'Joji Hirota Taiko Drummers': 1, 'AnnaMaria Bergqvist': 1, 'AWex': 1, 'Ge'La': 1, 'Hexell, Sixtho, Freyer': 1, 'Valery Teashler': 1, 'Ежемесячные': 1, 'Mike Champion, EarthQuake': 1, 'Amaya': 1, 'Chris Rea': 1, 'Banda La Fugitiva De Mike Miramontes': 1, 'Saison': 1, 'Angaleena Presley': 1, '天音': 1, 'Yoshiyuki Osawa': 1, 'Hot Chip': 1, 'YUZU, SEKAI NO OWAR I': 1, 'Rev. John Vijey': 1, 'Хор Сретенского монастыря': 1, 'Elvy Sukaesih': 1, 'Waa Wei': 1, 'Ana Popovic': 1, 'Natal': 1, 'Karen Davis': 1, 'Kotiteollisuus': 1, 'Giacinto Schiatti, Raffaele Trevisani, Ensemble Barocco Carlo Antonio Marino, Natale Arnoldi': 1, 'Antoine Forqueray, Atsushi Sakai, Christophe Rousset, Marion Martineau, Isabelle Saint-Yves': 1, 'Subalpine': 1, 'Los Violines de Villafontana': 1, 'Mezzo Piano': 1, 'RW l est temps de dormir profondément': 1, 'Fucking Werewolf Asso': 1, 'Helena Vondráčková': 1, 'Minimusicman': 1, 'Berceuse Academie': 1, 'Red Dirt': 1, 'Mahasneh': 1, 'Sentya Sapulette': 1, 'Ероп и Наталия Лансепе': 1, 'George Frideric Handel, Josep Maria Martí Duran': 1, 'Lang Lang, Andra Day': 1, 'K. S. Harisankar, Shreya Ghoshal, Kailas': 1, 'daigoro789': 1, 'Spin Doctors': 1, 'Orlande de Lassus, Die Singphoniker': 1, 'Saccao, Gorkiz': 1, 'DJ Tira, Black Motion, Drumetic Boyz': 1, 'Zwe Htet': 1, 'Amanda Larson': 1, 'Johan Daniel Berlin, Gottfried Von Der Goltz, Norwegian Baroque Orchestra': 1, '中山美穂': 1, '鄭興': 1, 'Francisco Tárrega, Giulio Tampalini': 1, 'Paula Menga': 1, 'Mickey Cora Y Su Orquesta Cábalá': 1, 'Pichy Valdés, Saoco Son': 1, '陳永淘': 1, 'Re-TROS, Liars': 1, 'Beta 2, Steo': 1, 'Windhand': 1, 'Luis Miguel Del Amargue': 1, 'Allan Taylor, Michael Koschorrek, Ralf Gustke, Hans-Joerg Maucksch, Dieter Burmester, Michael Kleinhans': 1, 'Johann Sebastian Bach, Nicholas Danby': 1, 'Calibre': 1, 'Novelbright': 1, 'Alexander Broussard': 1, 'Wilki': 1, 'Johann Sebastian Bach, English Baroque Soloists, John Eliot Gardiner': 1, 'Yuki Ijuin, CARREC': 1, 'Pongsit Kampee': 1, 'The Vintage Vandals': 1, 'Camilla Sparksss': 1, 'Christmas Hits Collective': 1, 'Christine D'Clario': 1, 'Suncatcher': 1, 'Murat Yalçın': 1, 'MI GYO': 1, 'Tim Dian': 1, 'Pure Obsessions & Red Nights, An Erotic End of Times': 1, 'Fariborz Lachini': 1, 'LaCrème, Ladybird': 1, 'Manatsu Nagahara': 1, 'Fox Amoore': 1, 'Leo Brouwer, Sharon Isbin': 1, 'Alex Verdugo': 1, 'Lazuli': 1, 'Hoài Lâm': 1, 'Pete Bellis & Tommy, Costa Mee': 1, 'Matt Doyle': 1, 'Relaxing Mindfulness Meditation Relaxation Maestro': 1, 'Ringbaan': 1, 'Kibariye': 1, 'Bernard Allison': 1, 'Aretha Franklin, Papercha\$Er': 1, 'Jacksonifyer': 1, 'Bouzouki Kings, Giorgos Zambetas': 1, 'Kiki Doll, Luca Debonaire': 1, 'Massage Therapy Studio': 1, 'Sweet Charity': 1, 'Gáspár Kálmán Leon, Khamoro Budapest Band': 1, 'Solomon Grey': 1, 'Gammalgrass': 1, 'Uptown Vocal Jazz Quartet, Richie Cole': 1, 'Nicu Măță, Paul Ananie, Gabriel Dumitru, Orchestra Lăutarii': 1, 'Orien': 1, 'Roy y Los Gentiles O

ficial': 1, 'NNS': 1, 'Khaled Abdrabo': 1, 'Marc Hartman': 1, 'Joris Voorn', 'Kid A': 1, 'Celtas Cortos, Nomadi': 1, '同心圓·敬拜者使團TWS': 1, 'Jeff Pi erre': 1, 'Sammi Cheng, David Ling Jr': 1, 'Nao Matsushita': 1, 'Orchestre de la fin du monde': 1, 'Chad Michael Butler, Yuna Kim': 1, 'Bi-2': 1, '遠見真矢(CV:松本まりか)': 1, 'Elder Anthony Boachie, Great Ampong': 1, 'Hilary Woods': 1, 'Shams, Muboraksho Mirzoshoev': 1, 'David Tavares, André Ribas': 1, 'Stephanie Erastus': 1, 'Buleria': 1, 'Cezinando, Chirag': 1, 'Darkth rone': 1, 'DJ Antoine, Armando, Jimmi The Dealer, Groove81': 1, 'Trygve Sk aug': 1, 'Sinister Skeletons': 1, 'Scheherazade': 1, 'Arnaud Rebotini': 1, 'Evgeny Nazarenko': 1, 'Olavi Uusivirta': 1, 'Coakira': 1, 'Dawn Tyler Wat son, Ben Racine': 1, 'Daniele De Martino, Valentina Belli': 1, 'Timothy Ja mes Bowen': 1, 'Ale Möller, Lena Willemark, Hans Ek, Västerås Sinfonietta, Rafael Sida Huizar': 1, 'Riverdale Cast, Camila Mendes, Cole Sprouse, KJ A pa, Lili Reinhart': 1, 'Accuface': 1, 'Lucinda Williams': 1, 'Noemi Huamá n': 1, 'KLANGSTEIN': 1, 'Marno ASMR': 1, 'Mark Barrott': 1, 'Đan Trường': 1, 'Carlos Gallardo & Peyton': 1, 'Борис Акунин': 1, 'Kjell Inge Torgersen ': 1, 'The Haggis Horns, Doc Brown': 1, 'Petrona Gonzales': 1, 'Neno Belan , Fiumens': 1, 'Keni Rightout': 1, 'Honey Colonna, Highway Superstar': 1, 'Dorsaf Hamdani': 1, 'Simfònica de Cobla i Corda de Catalunya, Manu Guix, Elena Gadel': 1, 'ASMR Glow': 1, 'İbrahim Can': 1, 'Chandimal Fernando, Ma hi Pasquel': 1, 'Teresa Parodi': 1, 'Diana Gabaldon, Birgitta Assheuer': 1 , 'Hirai Zerdüş': 1, 'El Remolón': 1, 'Dore': 1, 'Retief Burger': 1, 'Yuki ko Okada': 1, 'Dave Gahan': 1, 'Lluís Coloma Trio, Lluís Coloma, Manolo Ge rmán, Marc Ruiz': 1, 'Electric Dragon': 1, '西木野真姫(CV.Pile)': 1, 'Vagin a Lips': 1, 'Jacana Jacana': 1, 'Radnor': 1, 'EAMONN KARRAN': 1, 'Dorel de la Popesti': 1, 'Pilot': 1, 'Haleiwa': 1, 'Ngọc Hân': 1, 'Kobosil': 1, '阿 尾茂毅': 1, 'Daniel Aguayo, Regan': 1, 'Marcio Teixeira, Coral Voice Soul': 1, 'Brandon Echavarria, legado, Solangie Rodriguez, Alto Destino, Andrés G uerrero, Angie Ramirez, Antonella Guerrero, Carlos Hurtado, Leydy Ulcue, M aria Camila Novoa, Sara Novoa, Valery Guerrero': 1, 'Compay Segundo': 1, 'The Synthphonix': 1, 'Füstös': 1, 'Ato Libertário': 1, 'Yamaoka': 1, 'Hayd ée Milanés, Kelvis Ochoa': 1, 'Cotton Claw': 1, '玻璃海樂團, Airy Chan, Gab by Yeung': 1, 'Dope D.O.D.': 1, 'Ада': 1, 'Julio Elias': 1, 'Arno Strobel, Sascha Rotermund': 1, 'Alfa Sekta': 1, 'MoshTekk': 1, 'Ventre': 1, 'Miltat ek': 1, 'Mashina Vremeni': 1, 'Acoustic Heartstrings': 1, 'Marc Anthony': 1, "James Brown, The J.B.'s": 1, 'Andreas Waldetoft': 1, 'Zion Worship, Pa uli Salas': 1, 'Low Deep T': 1, 'Ty Ardis, Albert Lennard Project': 1, 'Gr uselkabinett, Michael Pan, Matthias Lühn, Horst Naumann, Claudia Urbschat- Mingues, Hanns Heinz Ewers': 1, 'Workout Dance Factory': 1, 'Dexter Ward': 1, 'Sideform, Sonic Entity': 1, 'Ana Consejos, Musica Relajante': 1, 'Kraf ty Kuts, Dynamite MC, Chali 2na, Kraak & Smaak': 1, 'Mozes and the Firstbo rn': 1, 'Dana Gillespie': 1, 'Ylli Demaj, Adelina Tahiri, Ryva Kajtazi': 1 , 'Mors Principium Est': 1, 'Yuuki Okamoto': 1, 'Terri Geisel': 1, 'Michae l Dease, Tim Green, Glenn Zaleski, Rodney Whitaker, Ulysses Owens': 1, 'En tspannungszeit': 1, 'Chimpo, DRS, Slay, Rago Loco': 1, 'The Sleeping Tree, Niam': 1, 'Jeffrey Paradise': 1, 'Ungdomskulen': 1, '歌っちゃ王': 1, 'Adoro ': 1, 'Sandy Turnbull': 1, 'Search': 1, 'Counting Clouds': 1, 'Relaxing Or gel': 1, 'La Pitufobanda': 1, 'UNCHAIN': 1, 'Thu Uyên': 1, 'Mahesa': 1, 'C hiki Liki Tu-a': 1, 'Gregory Isaacs': 1, 'JayRex SuiSui, Jharvii, Robby T': 1, 'Măeana': 1, 'Vanilla': 1, 'Re:Lilly': 1, 'Jana Rychterová': 1, 'Baby Band': 1, 'Mellina': 1})

In [8]: `# simple analyst`

```
sns.histplot(df['duration_(minutes)'],kde=True,color='green')
plt.title ('Frekuensi Duration Musik in sportify')
```

Out[8]: Text(0.5, 1.0, 'Frekuensi Duration Musik in sportify')



In [9]: `df.describe()`

Out[9]:

	duration_(minutes)
count	816.000000
mean	6.980392
std	5.907642
min	5.000000
25%	5.000000
50%	6.000000
75%	7.000000
max	100.000000

```
In [10]: # most artist has musik
most=count.most_common(5)
print ('top 5 artist')
name =[i for i,k in most ]
value=[k for i,k in most]
df_=pd.DataFrame({
    'artist':name,
    'total musik':value
})
df_
```

top 5 artist

Out[10]:

	artist	total musik
0	Nusrat Fateh Ali Khan	2
1	Tangerine Dream	2
2	Joachim Pastor	2
3	Suara	2
4	Nature Sounds	1